Chapter 5: Elasticity: Responsiveness of Demand and Supply

Principles of Microeconomics Curriculum

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Abstract

This chapter provides a rigorous, pedagogical treatment of Chapter 5: Elasticity: Responsiveness of Demand and Supply, benchmarked to standard principles curriculum.

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1 Midpoint Arc Elasticity Formula

$$\varepsilon_d = \frac{(Q_2 - Q_1)/[(Q_1 + Q_2)/2]}{(P_2 - P_1)/[(P_1 + P_2)/2]} \tag{1}$$

Demand is classified as elastic if $|\varepsilon_d| > 1$, unit elastic if $|\varepsilon_d| = 1$, and inelastic if $|\varepsilon_d| < 1$.

2 The Total Revenue Test

Total Revenue $TR = P \times Q$. Differentiating with respect to price yields:

$$\frac{d(TR)}{dP} = Q(1 - |\varepsilon_d|) \tag{2}$$

When demand is inelastic ($|\varepsilon_d| < 1$), price increases raise total revenue.